

Lumberton Mitigation Analysis

Mitigating the Deluge: A Resilience Approach for Lumberton, NC

CGN 5432 Risk and Reliability Applications in Natural Hazards

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Introduction



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Modeling Process



Results



Conclusion

Introduction

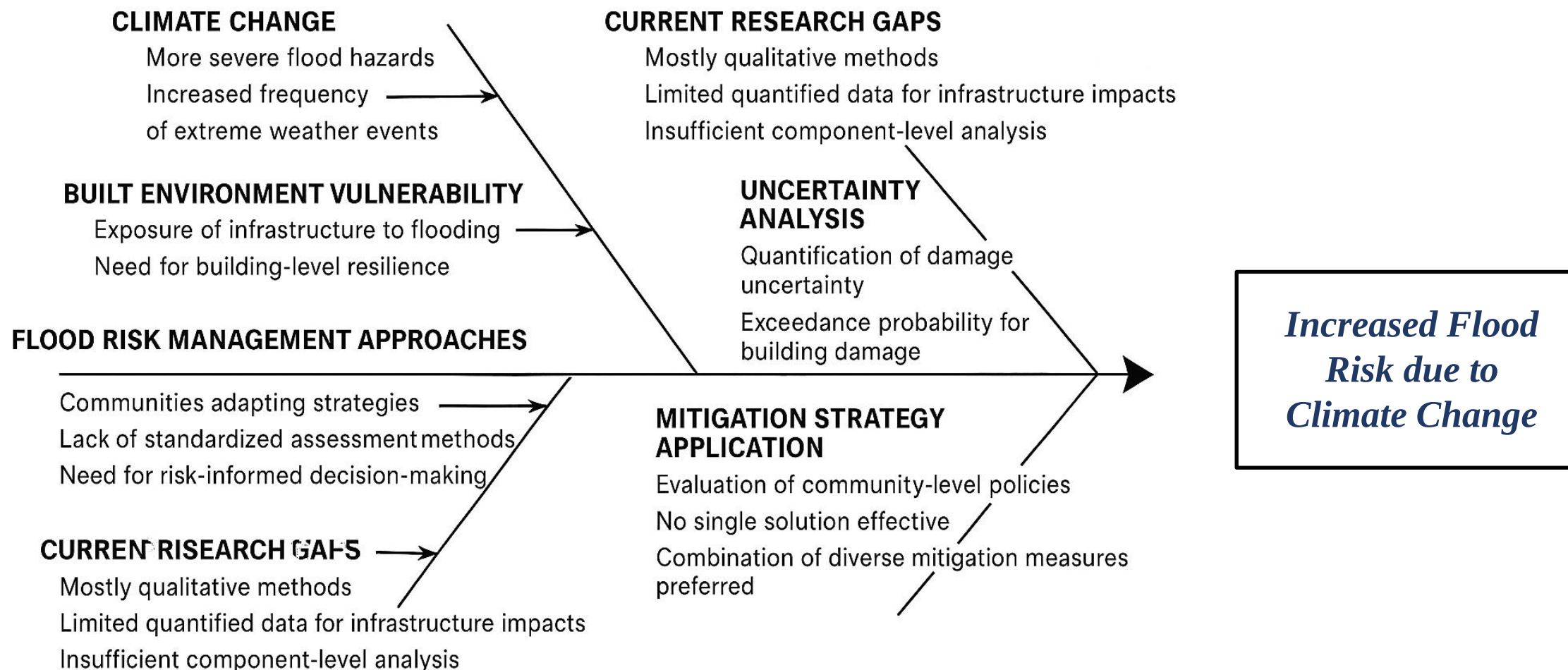
- ❑ Flooding is a critical hazard for communities like **Lumberton, NC**, requiring urgent mitigation planning.
- ❑ This project focuses on analyzing **flood risk and reliability** through advanced modeling techniques.
- ❑ Key emphasis is placed on **building damage assessments** using fragility curves across multiple archetypes.
- ❑ The study explores a variety of structural and non-structural mitigation strategies, including **flood barriers, pumps, and elevating components**.
- ❑ Simulation models are used to assess the effectiveness of each mitigation scenario under different hazard intensities.
- ❑ The goal is to identify **cost-effective, high-impact mitigation solutions** to enhance resilience and reduce future flood losses.



Figure 1-1. Path and intensity timeline of Hurricane Matthew.

Figure References: Sutley, E. J., Dillard, M. K., & van de Lindt, J. W. (2021). *Community resilience-focused technical investigation of the 2016 Lumberton, North Carolina flood* : <https://doi.org/10.6028/NIST.SP.1230-2>

IMPACT OF CLIMATE CHANGE ON FLOOD RISK



Nofal, O. M., & van de Lindt, J. W. (2021). High-resolution flood risk approach to quantify the impact of policy change on flood losses at community-level. *International Journal of Disaster Risk Reduction*, 62.

<https://doi.org/10.1016/j.ijdrr.2021.102429>

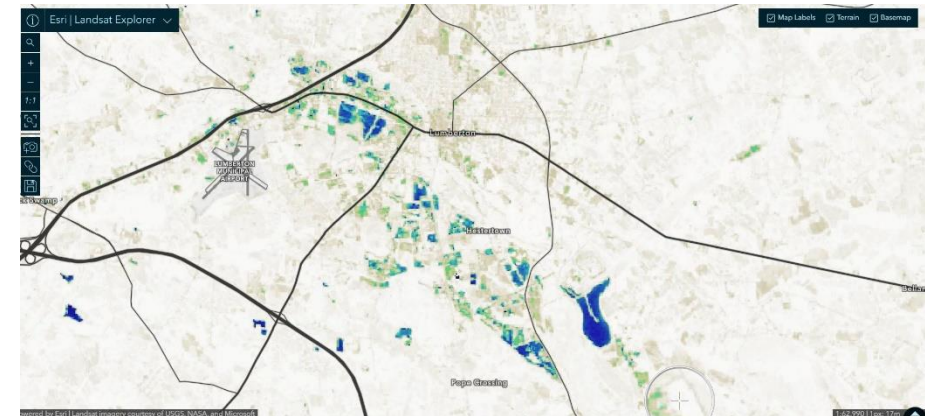
Objectives

- ❑ To assess flood-induced structural vulnerability using detailed building archetypes.
- ❑ To develop and analyze fragility curves representing different damage states across flood depths.
- ❑ To evaluate the effectiveness of various flood mitigation strategies, including:
 - ❖ **Flood Barriers**
 - ❖ **Pumps**
 - ❖ **Elevating Building Components**
 - ❖ **Combined Mitigation Scenarios**
- ❑ To simulate multiple hazard intensities and building responses using risk and reliability models.
- ❑ To identify optimal mitigation scenarios that reduce potential losses and enhance building resilience.



Why Lumberton, NC?

- ❑ Lumberton, located in **Robeson County, NC**, is historically flood-prone due to its proximity to the Lumber River.
- ❑ The city has suffered severe flooding in recent decades, especially during **Hurricanes Matthew (2016)** and **Florence (2018)**.
- ❑ **Hurricane Matthew** caused catastrophic damage, displacing thousands and inundating critical infrastructure.
- ❑ Due to topographic vulnerability and high percentages of low-income and minority populations, Lumberton is identified as a socially vulnerable community.
- ❑ These conditions make Lumberton an ideal case study for evaluating the impact of building-level mitigation strategies.
- ❑ Ongoing flood mitigation efforts in Lumberton by **FEMA** and local government make it a timely and relevant study location.
- ❑ The city's diverse building inventory provides a realistic and complex testbed for risk modeling.

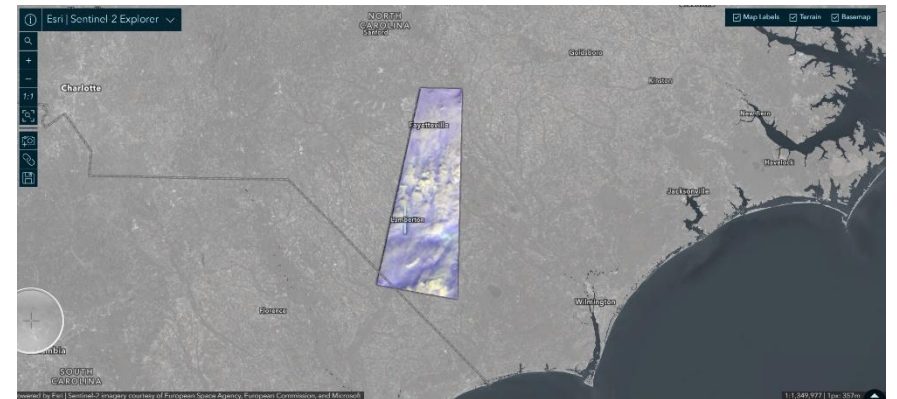


<i>FLOOD TYPE</i>	<i>DESCRIPTION</i>	<i>PROTECTIONS AGAINST FLOOD TYPE</i>
<i>River</i>	<i>Flooding resulting from the overflow of rivers or streams due to heavy rainfall or snowmelt inundates adjacent land areas.</i>	<i>Floodwall, Levee, Dikes, Dams</i>
<i>Precipitation</i>	<i>Flooding caused by excessive rainfall leads to surface water accumulation in urban or rural areas.</i>	<i>Floodwall, Levee</i>
<i>Coastal</i>	<i>Flooding occurs when seawater inundates coastal areas, typically due to storm surges, high tides, or sea-level rise.</i>	<i>Seawalls, Living breakwaters</i>

Ref: https://firststreet.org/county/robeson-county-northcarolina/37155_fsid/flood?from=riskfactor.com

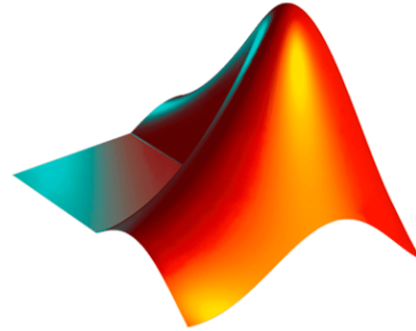
Modeling Process

- ❑ Developed a detailed flood risk assessment framework using probabilistic modeling.
- ❑ Identified and classified **15 distinct building archetypes** based on construction characteristics and vulnerability.
- ❑ Generated fragility curves for each archetype, covering damage states from minor (DS0) to complete damage (DS4).
- ❑ Incorporated **historical flood data**, depth-damage functions, and building inventory to calibrate the model.
- ❑ Simulated multiple flood scenarios with varying hazard intensities and durations.
- ❑ Applied **Monte Carlo simulations** to assess the probability of loss across different mitigation setups.
- ❑ Modeled mitigation strategies (**barriers, pumps, elevation**) individually and in combination.
- ❑ Compared results based on percentage damage reduction and cost-effectiveness.





ArcGIS Pro



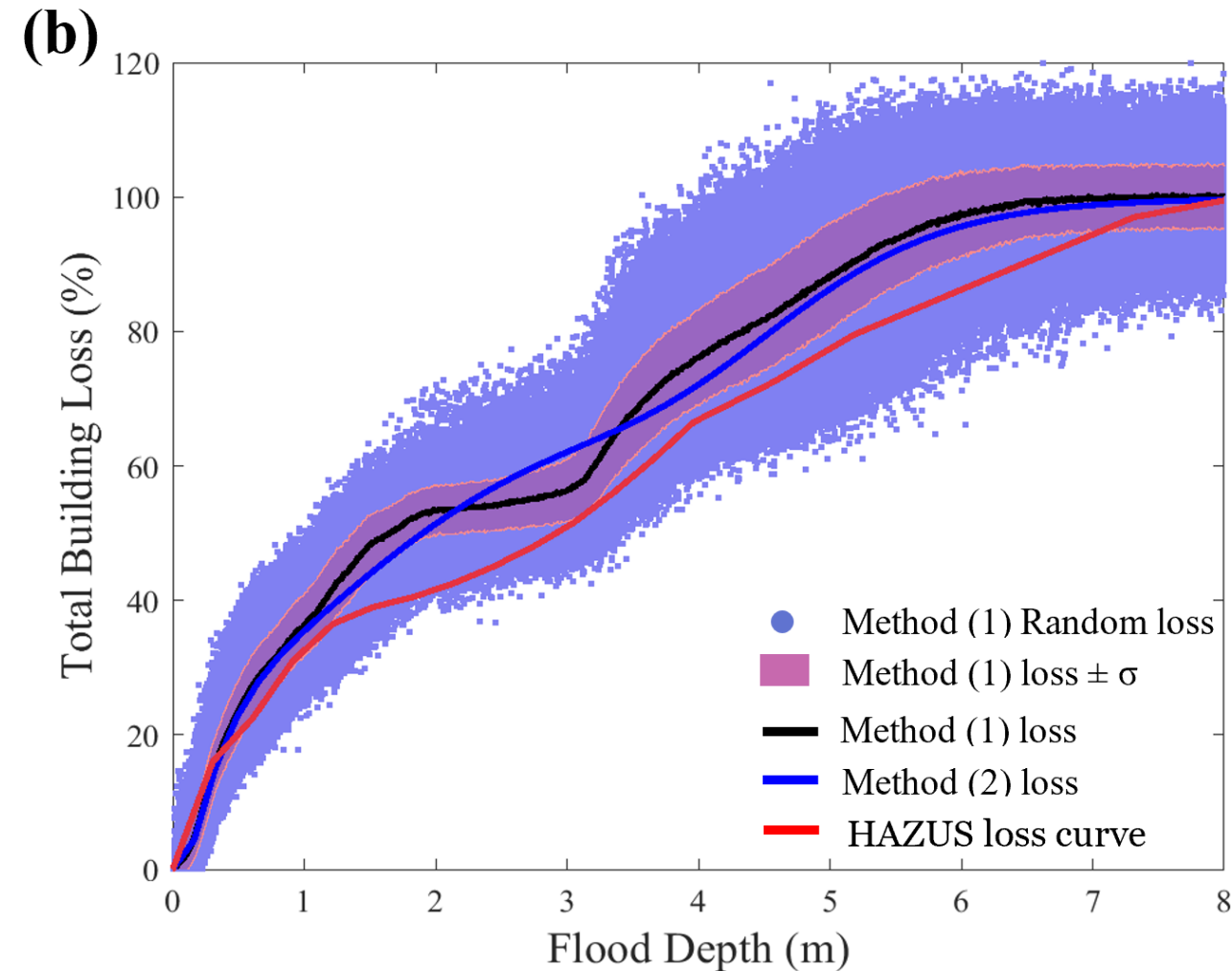
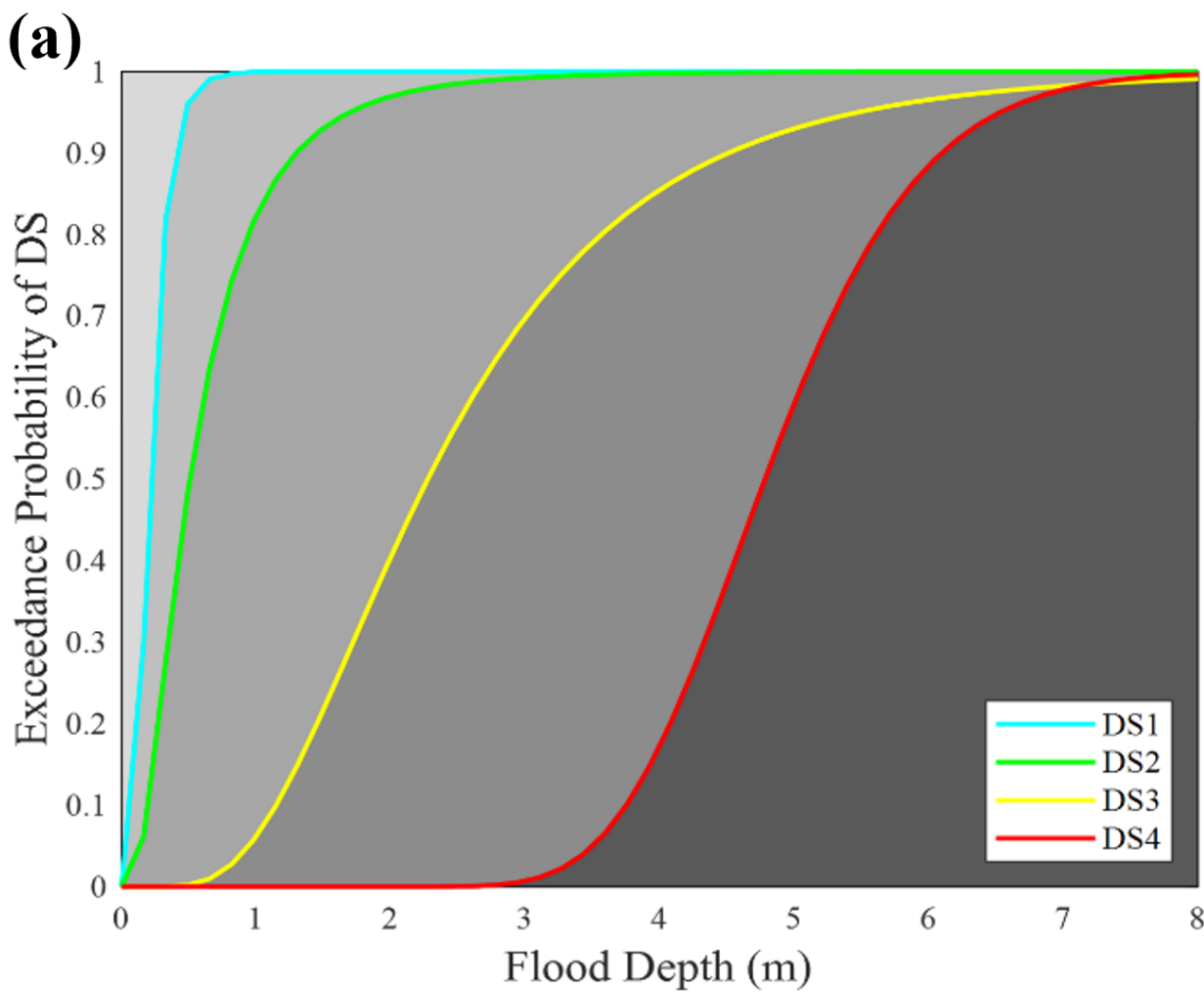
MATLAB®

Damage States

DS Level	Functionality	Description	Damage Scale	Damage Ratio
0	Operational	Insignificant damage to components below first-floor elevation. Minor damage to garage interiors including drywall, cabinets, electrical outlets, wall insulation. No sewer backup into the living area.	Insignificant	0.0 – 0.03
1	Limited Occupancy	Water touches floor joists up to minor water enters the building. Damage to carpets, pads, baseboards, flooring. No drywall damages with the potential of some mold on the subfloor above the crawlspace.	Slight	0.03 – 0.15
2	Restricted Occupancy	Partial damage to drywalls along with damage to electrical components (base-outlets), water heater, and furnace. Complete damage to major equipment, appliances, and furniture on the first floor.	Moderate	0.15 – 0.5
3	Restricted Use	Damage to the non-structural components and interiors within the whole building including drywall damage to upper stories for multi-story buildings. Electrical switches and mid-outlets are destroyed.	Extensive	0.5 – 0.7
4	Restricted Entry	Significant structural damage present. Non-structural components and interiors are destroyed including all drywall, appliances, cabinets, furniture, etc.	Complete	0.7 – 1.0

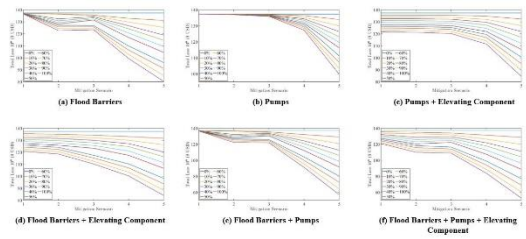
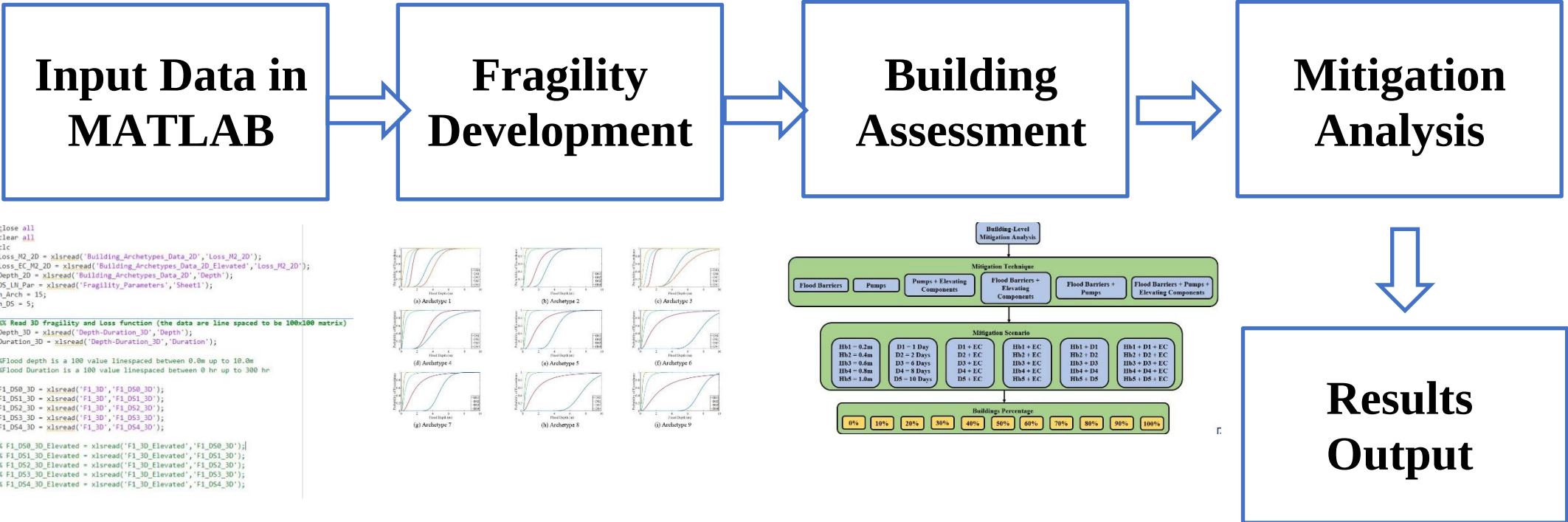
Residential Damage States and Income Levels

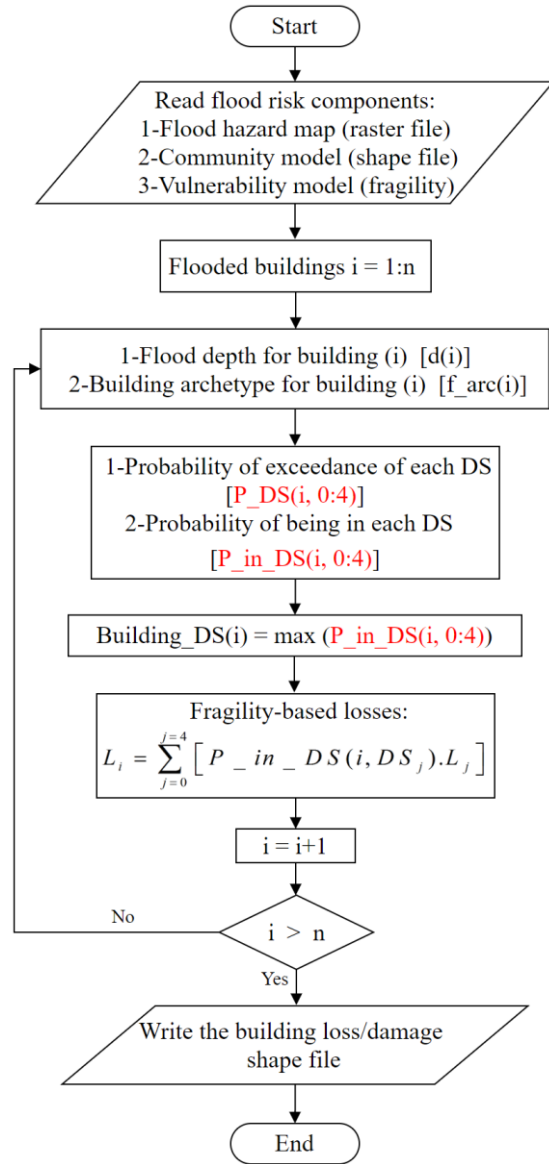
Damage State (DS)	Description
DS0	Insignificant damage to components below first-floor elevation . Water enters the crawlspace/basement and touches the foundation (crawlspace or slab on grade). Damage to components within the crawlspace/basement, including base insulation and stored inventory. Minor damage to garage interiors, including drywall, cabinets, electrical outlets, and wall insulation (Garage is below the first-flood elevation (FFE)). No sewer backup into the living area.
DS1	Water touches the floor joists, up to minor water entering the building. Damage to carpets, pads, baseboards, and flooring . Damage to the external AC unit (if the AC unit is not elevated) and the attached ductworks (if ductworks are in the crawlspace). Complete damage to the garage interior (if the garage is below FFE). No drywall damage with the potential of some mold on the subfloor above the crawl space. Could have a minor sewer backup and/or minor mold issue.
DS2	Partial damage to drywalls along with damage to electrical components (base-outlets), water heater, and furnace . Complete damage to major equipment, appliances, and furniture on the first floor. Damage to the lower bathroom and kitchen cabinets. Doors and windows may need replacement. It could have a major sewer backup and major mold issues.
DS3	Damage to the non-structural components and interiors within the whole building, including (but not limited to) drywall damage to upper stories for multi-story buildings (e.g., attic, second story, etc.). Electrical switches and mid-outlets are destroyed. Damage to bathroom/kitchen upper cabinets, lighting fixtures on walls, are destroyed with potential damage to ceiling lighting fixtures. Studs are reusable; some may be damaged. Major sewer backup will happen along with major mold issues. Equipment, appliances, and furniture on the upper floors are also damaged (e.g., attic, second floor, etc.).
DS4	Significant structural damage present (e.g., studs, trusses, joists , etc.). Non-structural components and interiors are destroyed, including all drywall, appliances, cabinets, furniture , etc. Damage to rooftop units/components, including roof insulation, sheathing, and electro-mechanical systems (rooftop AC units, electrical systems, cable railing, sound system, etc.). The foundation could be floated off. The building must be demolished or potentially replaced.



https://incore.ncsa.illinois.edu/doc/incore/notebooks/Lumberton_testbed/Lumberton_testbed.html

Nofal, O. M., & van de Lindt, J. W. (2021). High-resolution flood risk approach to quantify the impact of policy change on flood losses at community-level. *International Journal of Disaster Risk Reduction*, 62. <https://doi.org/10.1016/j.ijdrr.2021.102429>





Hazard, exposure, and vulnerability models were integrated into a community-level loss and damage analysis framework, facilitating the computation of flood-related losses and damages for each structure within the community, as seen in the flow chart below. This work is the inaugural use of a representative suite of flood fragilities for conducting community-level damage and loss assessment.

It is important to note that DS0 is exclusively applicable to archetypes with a crawlspace basis, namely archetypes F1 and F3. For all building archetypes, flood depth is assessed from the Finished Floor Elevation (FFE). For archetypes with crawl space foundations, flood depth is assessed from ground elevation, with the finished floor elevation (FFE) presumed to be 1.0m above ground level. This accounts for the deviation of DS1 and DS2 for archetypes F1 and F3 from the fragility suite.

Building Archetypes

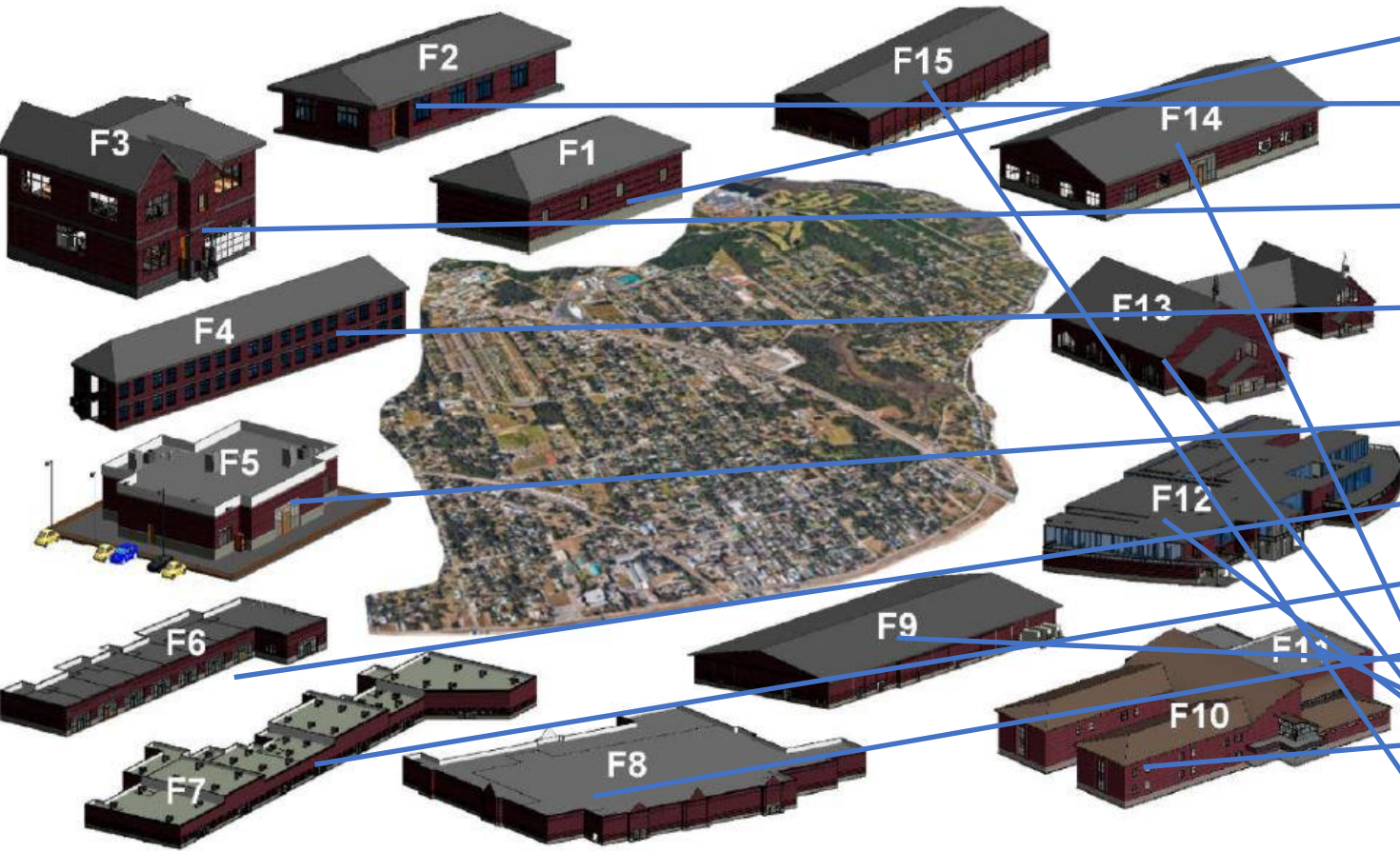
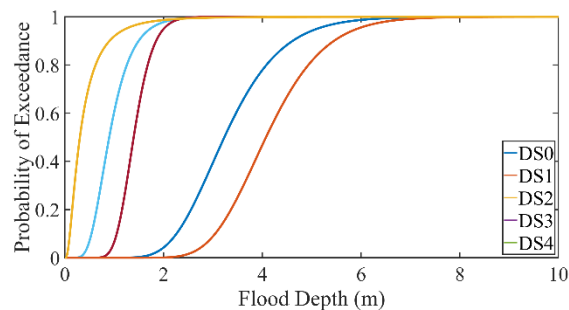
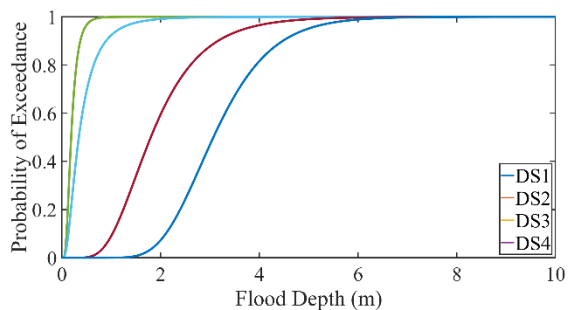


Figure 4. Schematic representation of using minimum building archetypes portfolio to model a community.

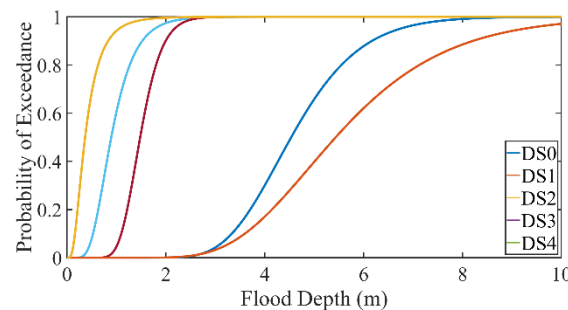
Building Archetype	Building Description
F1	One-story single-family residential building on a crawlspace foundation
F2	One-story multi-family residential building on a slab-on-grade foundation
F3	Two-story single-family residential building on a crawlspace foundation
F4	Two-story multi-family residential building on a slab-on-grade foundation
F5	Small grocery store/Gas station with a convenience store
F6	Multi-unit retail building (strip mall)
F7	Small multi-unit commercial building
F8	Super retail center
F9	Industrial building
F10	One-story school
F11	Two-story school
F12	Hospital/Clinic
F13	Community center (place of worship)
F14	Office building
F15	Warehouse (small/large box)



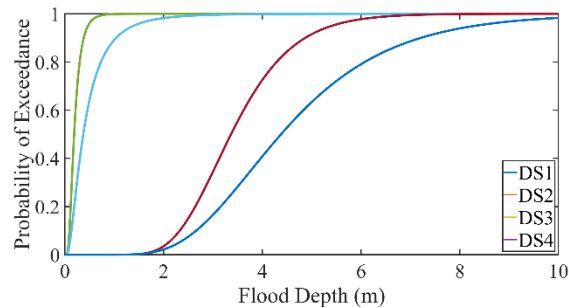
(a) Archetype 1



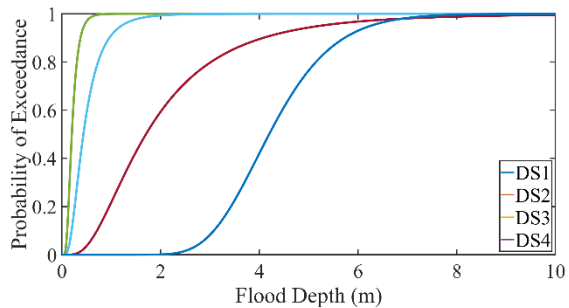
(b) Archetype 2



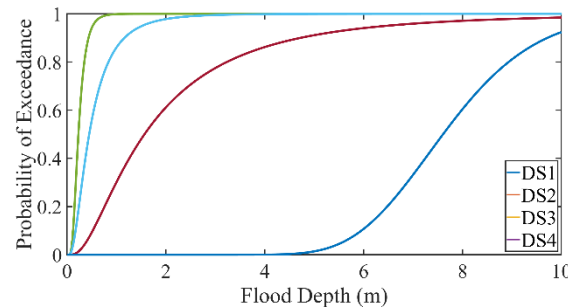
(c) Archetype 3



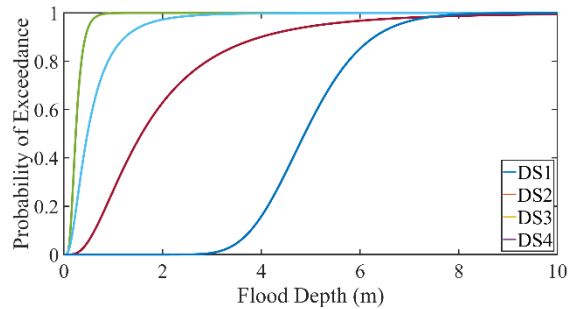
(d) Archetype 4



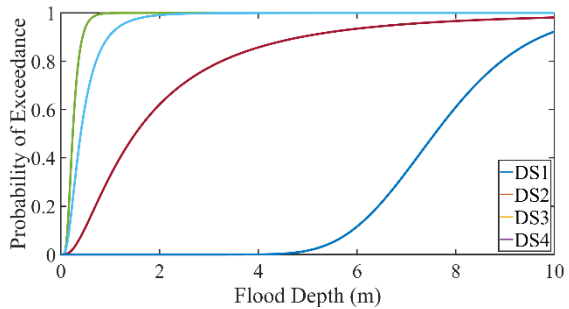
(e) Archetype 5



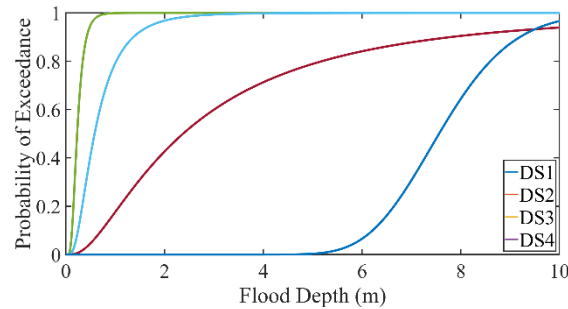
(f) Archetype 6



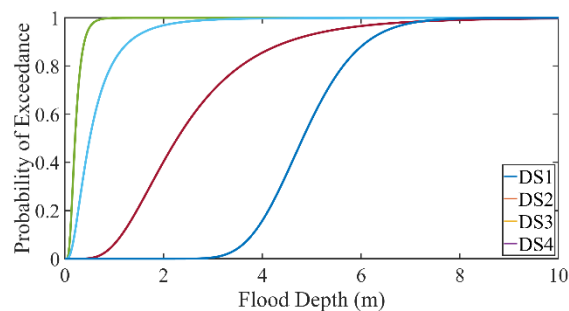
(g) Archetype 7



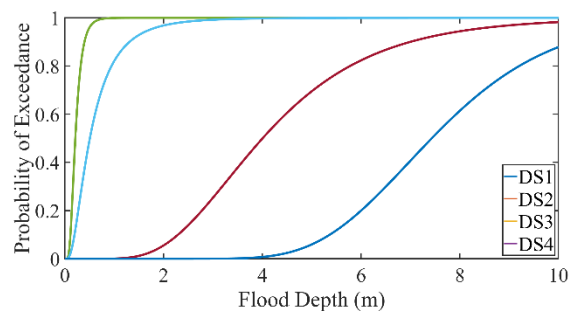
(h) Archetype 8



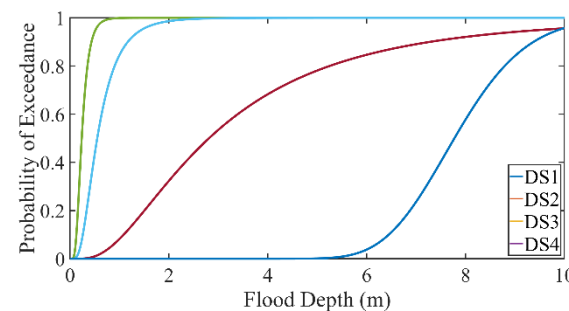
(i) Archetype 9



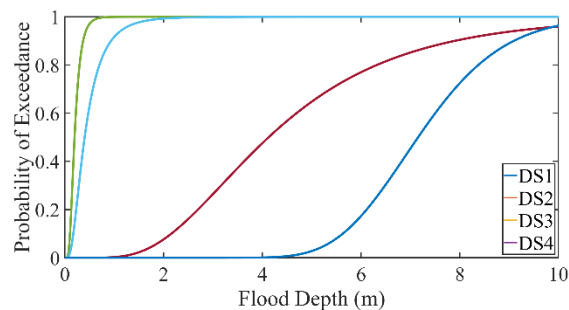
(j) Archetype 10



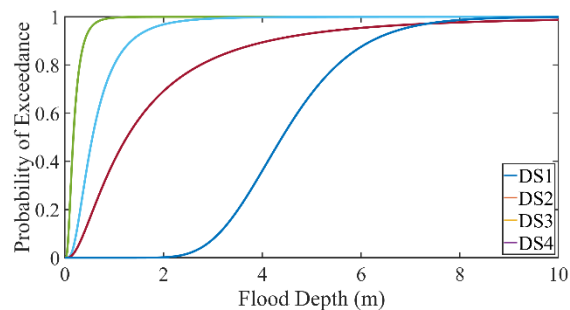
(k) Archetype 11



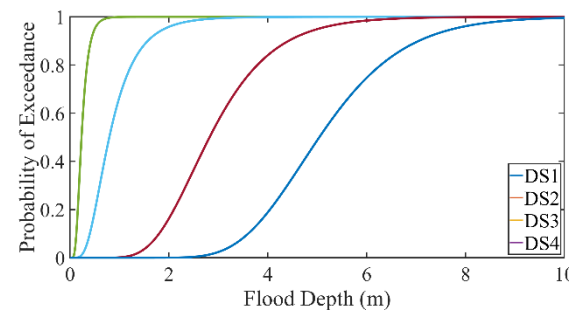
(l) Archetype 12



(m) Archetype 13

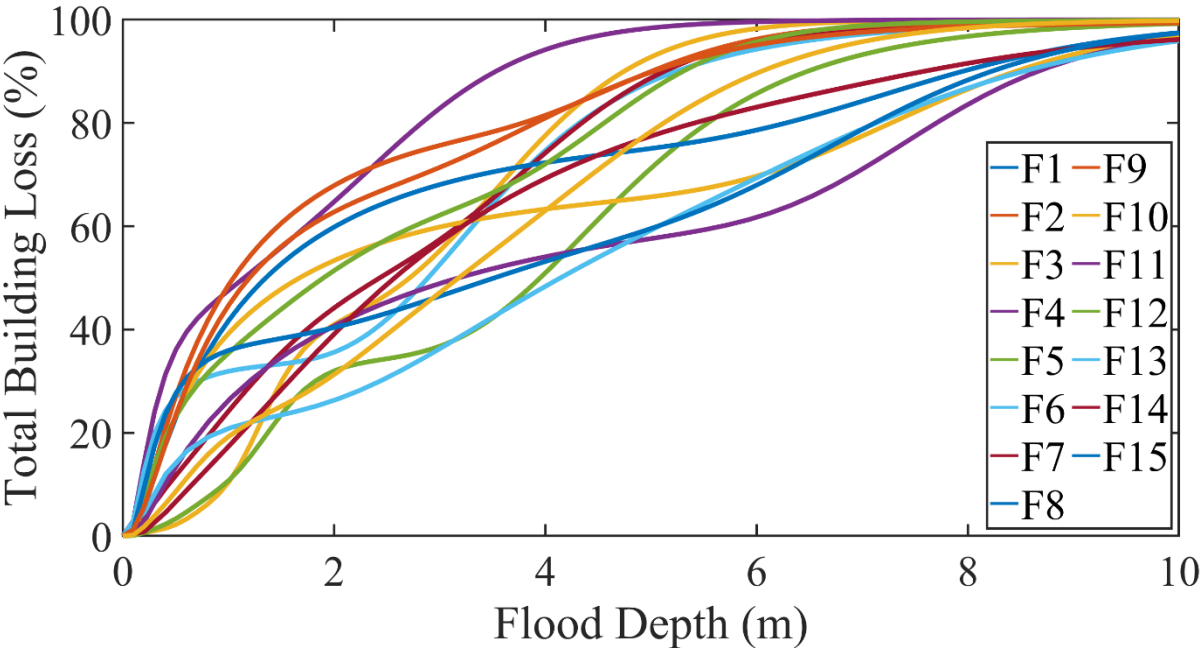
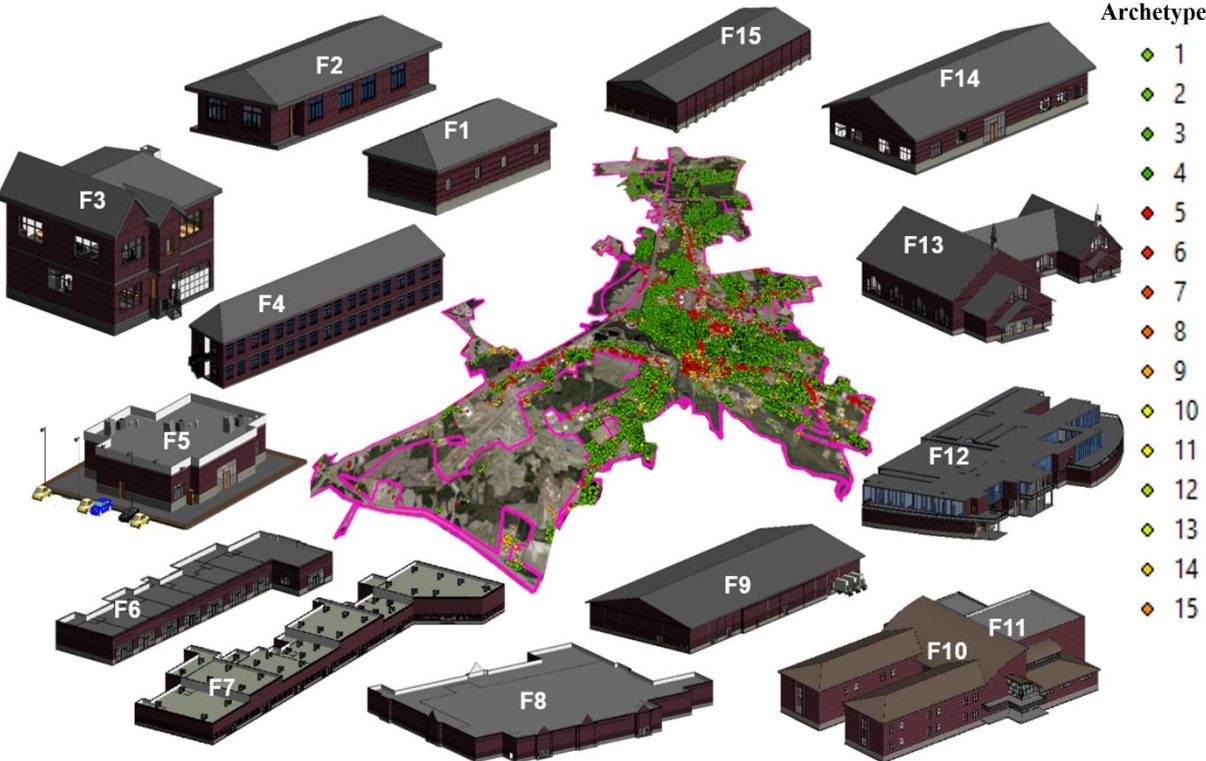


(n) Archetype 14

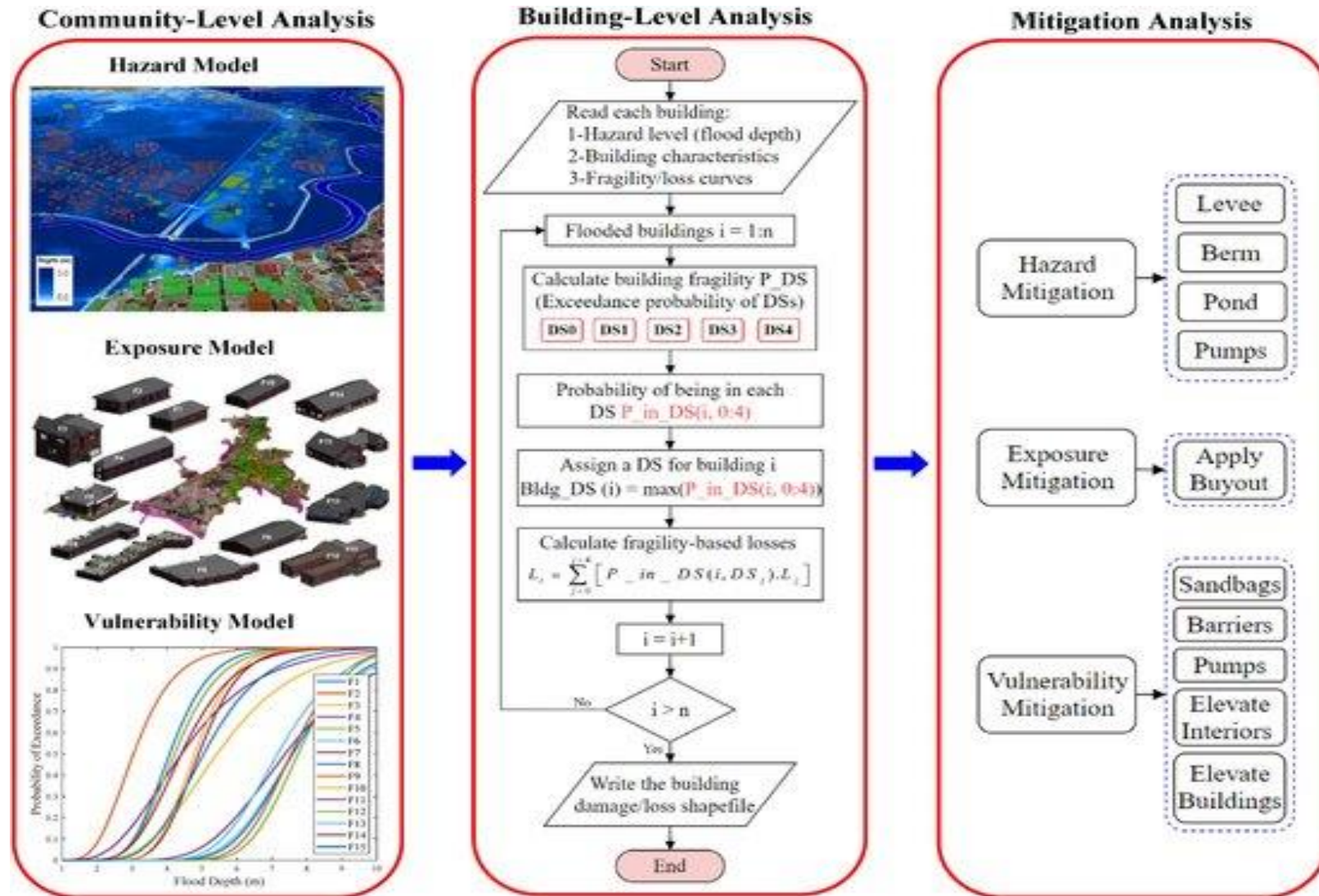


(o) Archetype 15

Archetypes Scenarios (combined)



Total building loss versus flood depth for all building archetypes.



Building-Level
Mitigation Analysis

Mitigation Technique

Flood Barriers

Pumps

Pumps + Elevating
Components

Flood Barriers +
Elevating
Components

Flood Barriers +
Pumps

Flood Barriers + Pumps +
Elevating Components

Mitigation Scenario

Hb1 = 0.2m
Hb2 = 0.4m
Hb3 = 0.6m
Hb4 = 0.8m
Hb5 = 1.0m

D1 = 1 Day
D2 = 2 Days
D3 = 6 Days
D4 = 8 Days
D5 = 10 Days

D1 + EC
D2 + EC
D3 + EC
D4 + EC
D5 + EC

Hb1 + EC
Hb2 + EC
Hb3 + EC
Hb4 + EC
Hb5 + EC

Hb1 + D1
Hb2 + D2
Hb3 + D3
Hb4 + D4
Hb5 + D5

Hb1 + D1 + EC
Hb2 + D2 + EC
Hb3 + D3 + EC
Hb4 + D4 + EC
Hb5 + D5 + EC

Buildings Percentage

0%

10%

20%

30%

40%

50%

60%

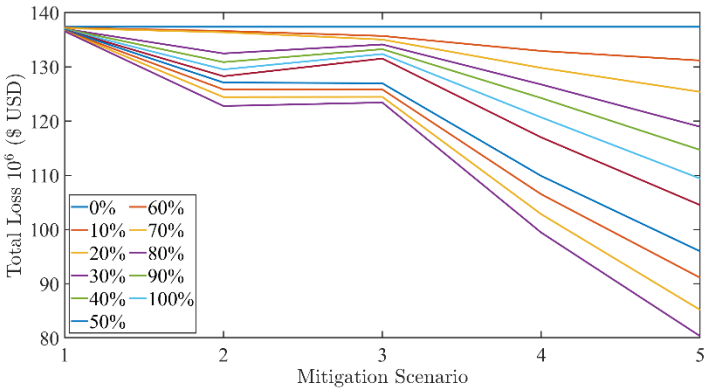
70%

80%

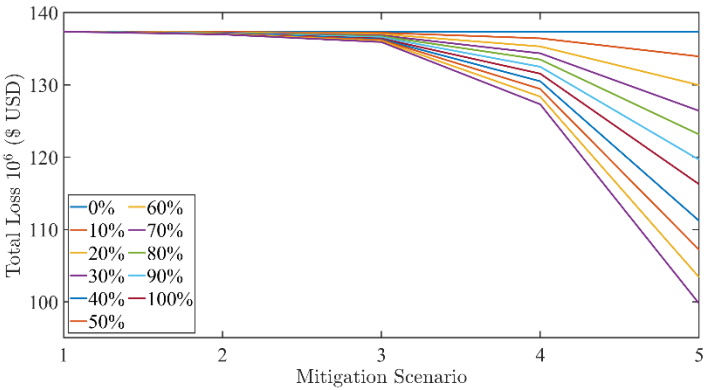
90%

100%

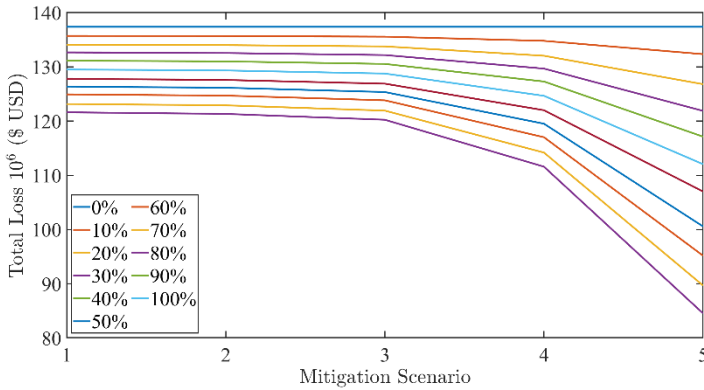
Mitigation Scenarios



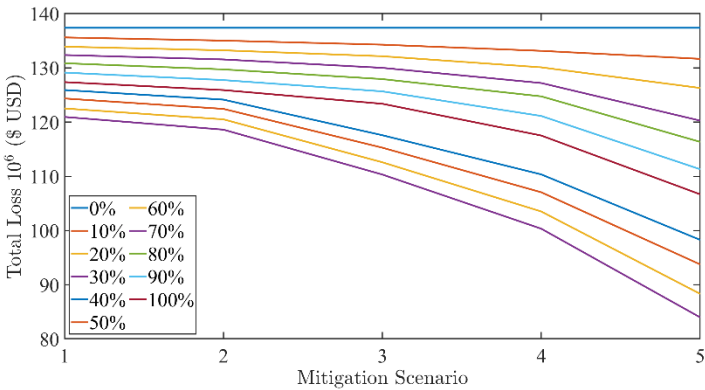
(a) Flood Barriers



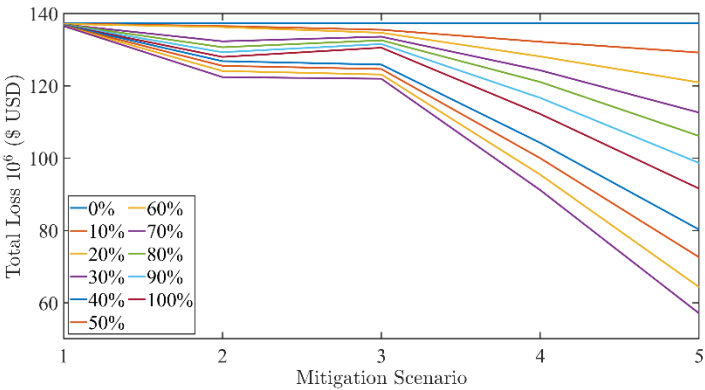
(b) Pumps



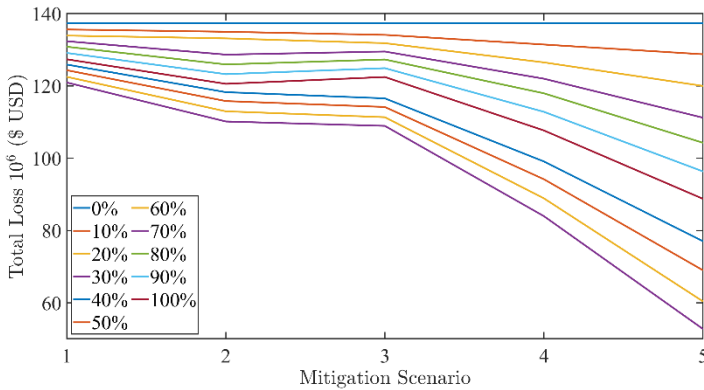
(c) Pumps + Elevating Component



(d) Flood Barriers + Elevating Component

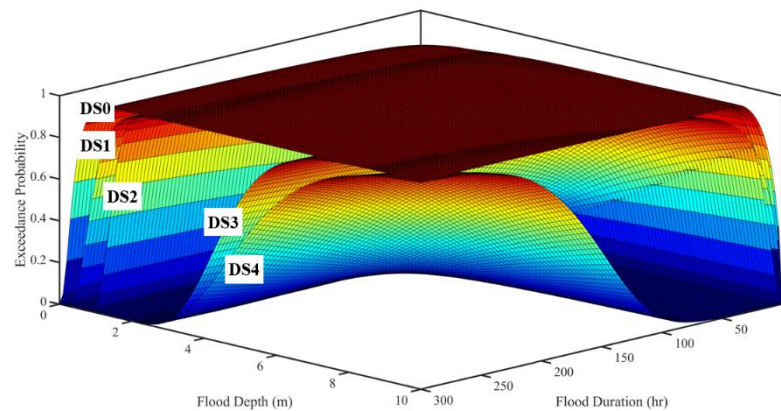


(e) Flood Barriers + Pumps

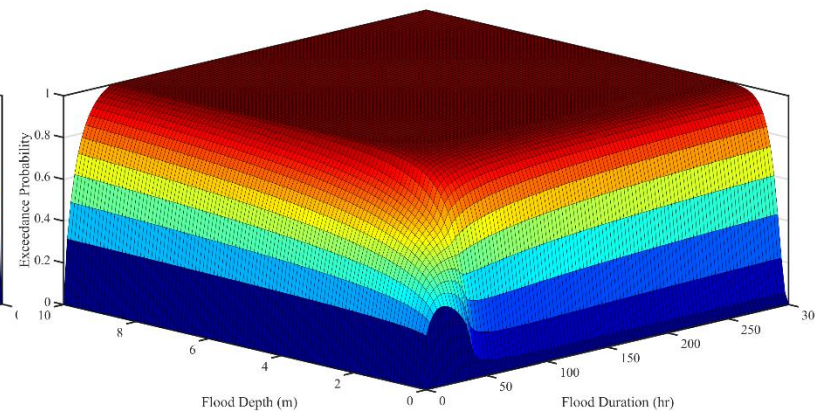


(f) Flood Barriers + Pumps + Elevating Component

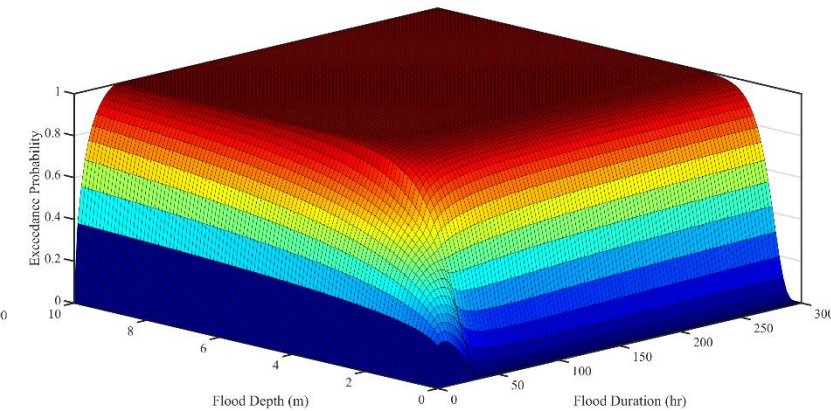
Mitigation Scenarios



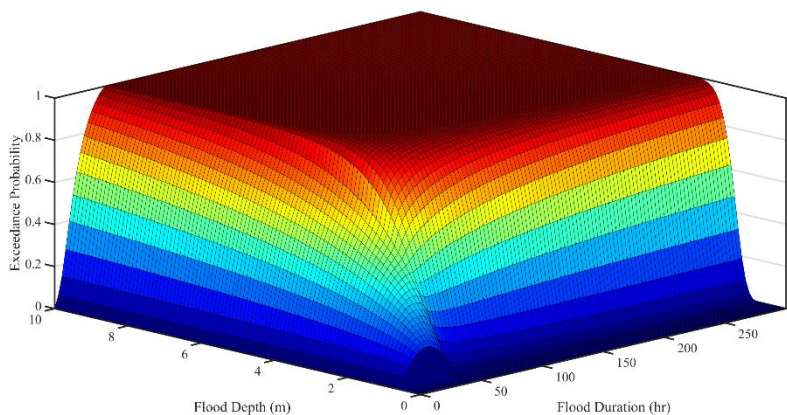
(a) All Building Fragility Curves



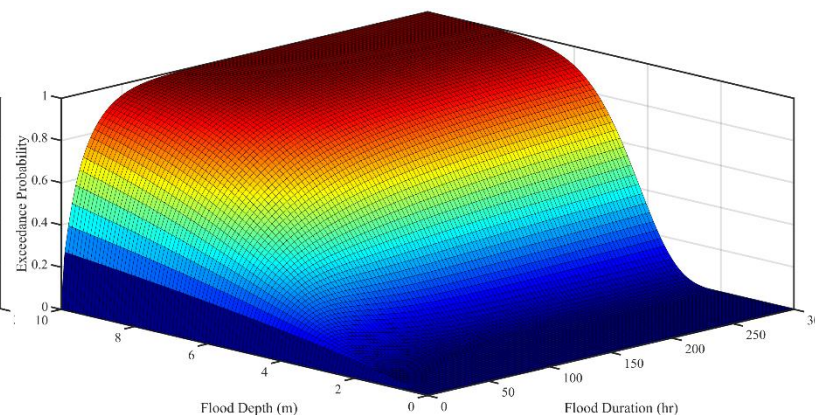
(b) Fragility Curve for DS0



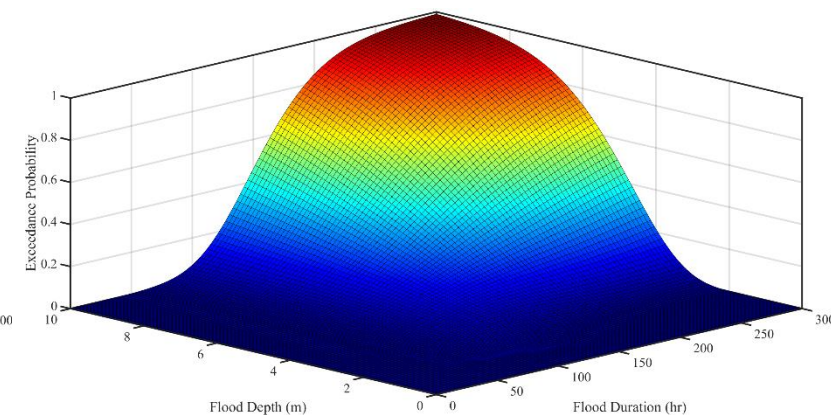
(c) Fragility Curve for DS1



(d) Fragility Curve for DS2

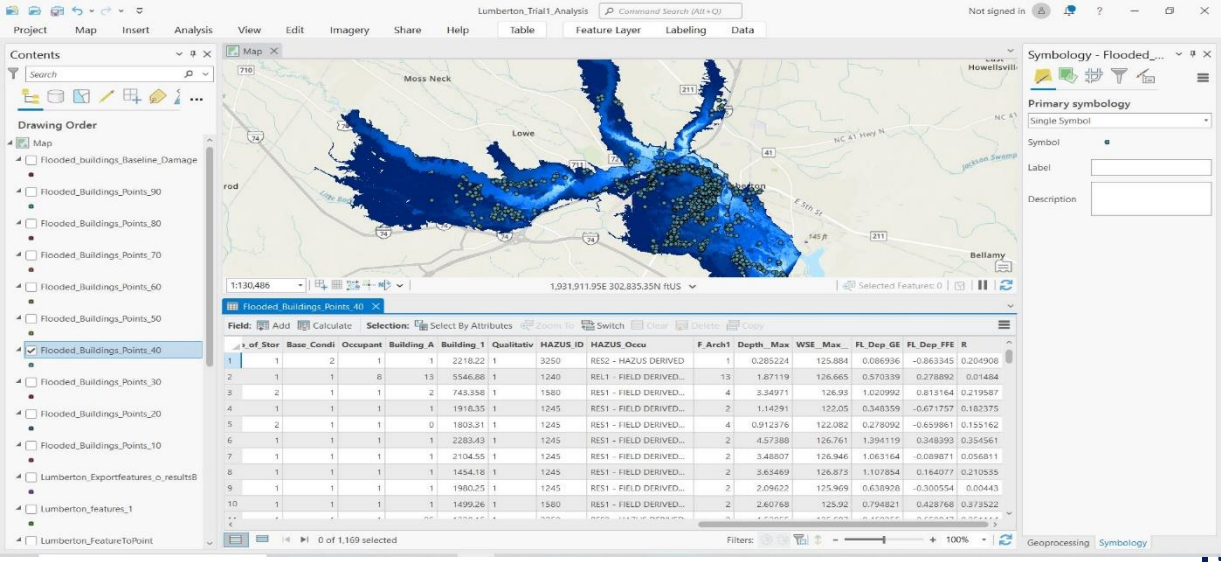
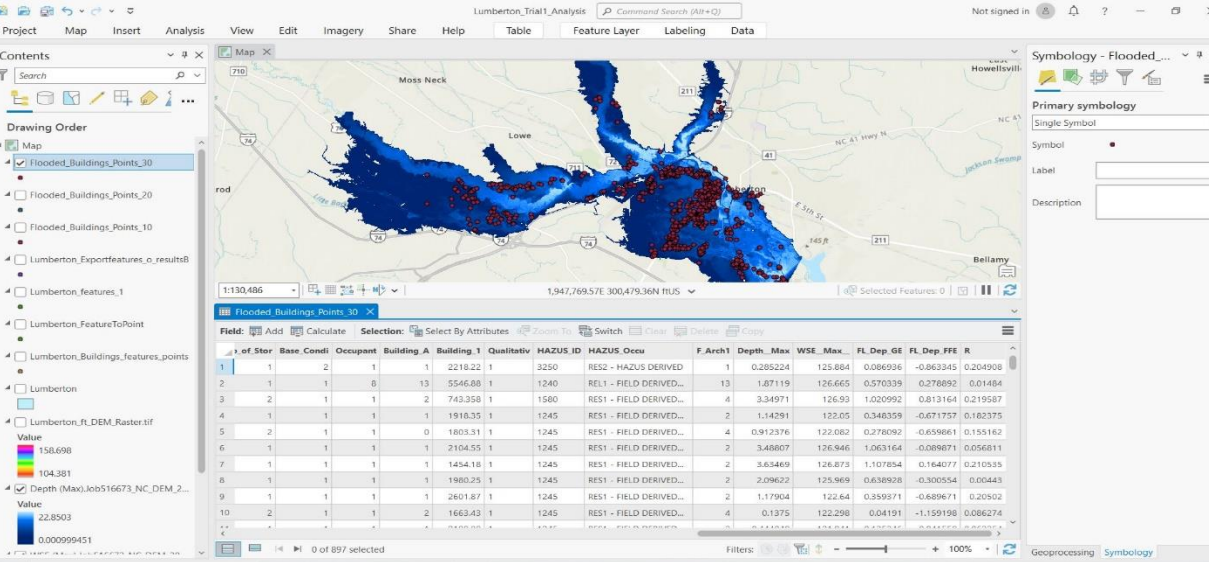
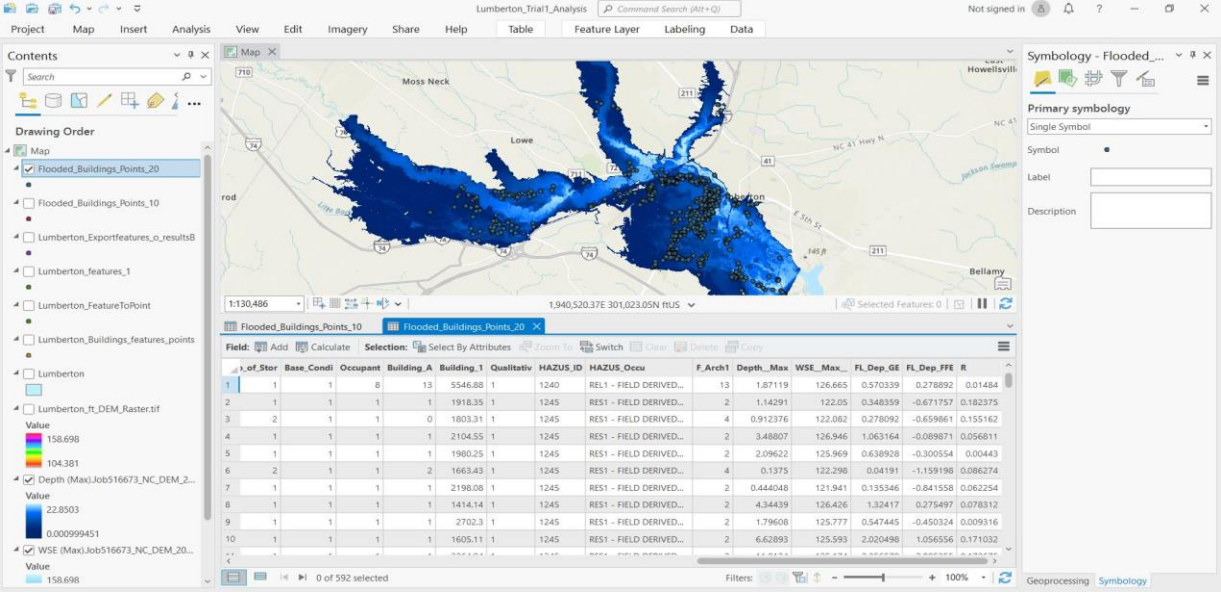
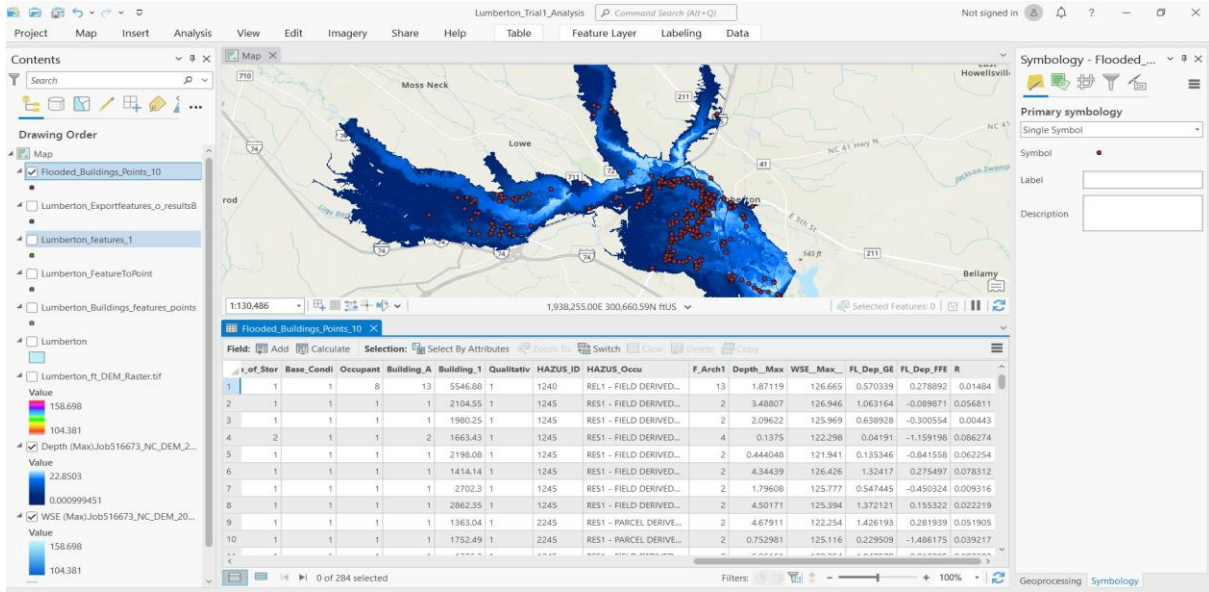


(e) Fragility Curve for DS3

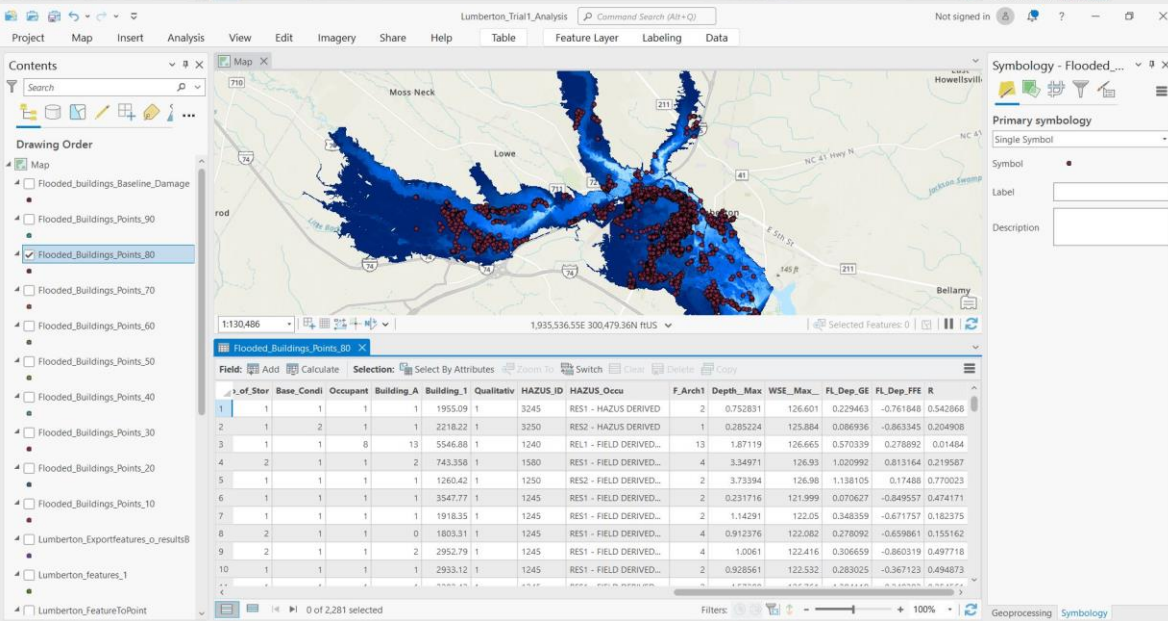
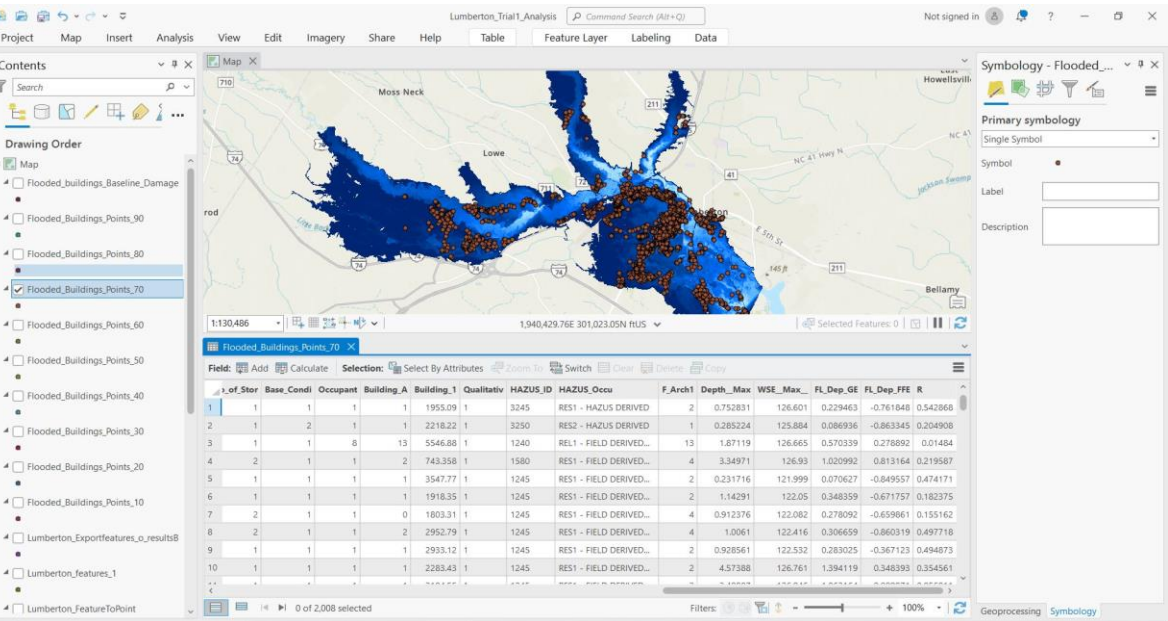
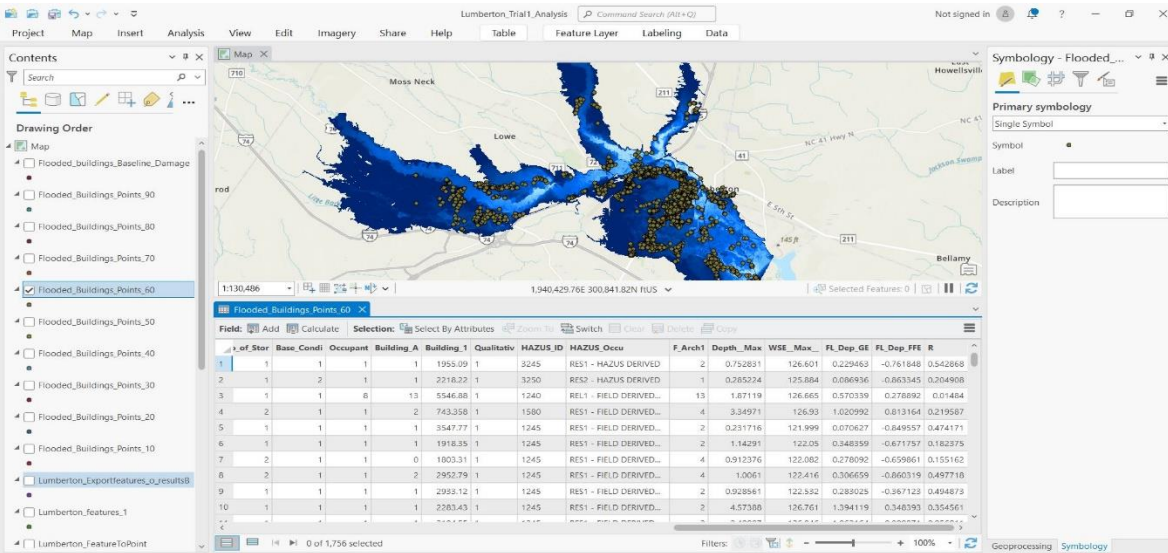
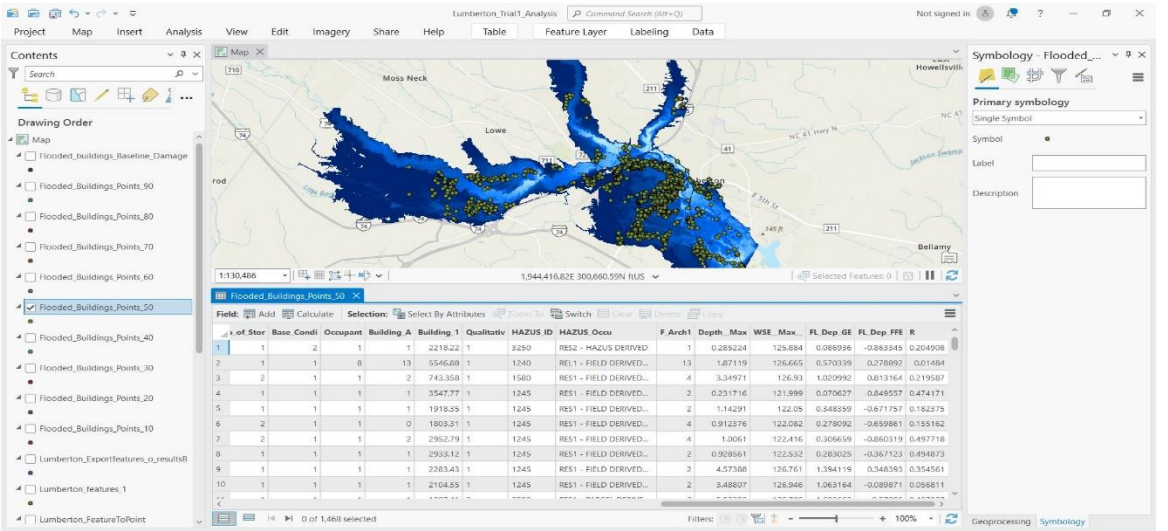


(f) Fragility Curve for DS4

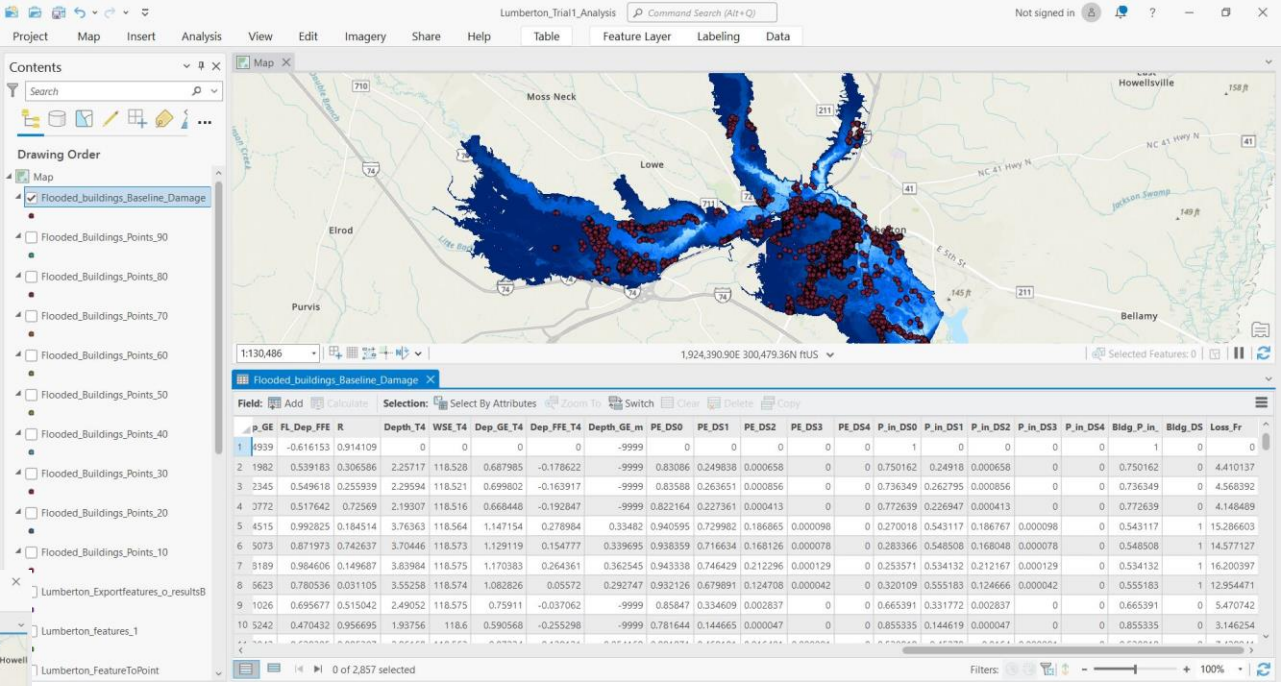
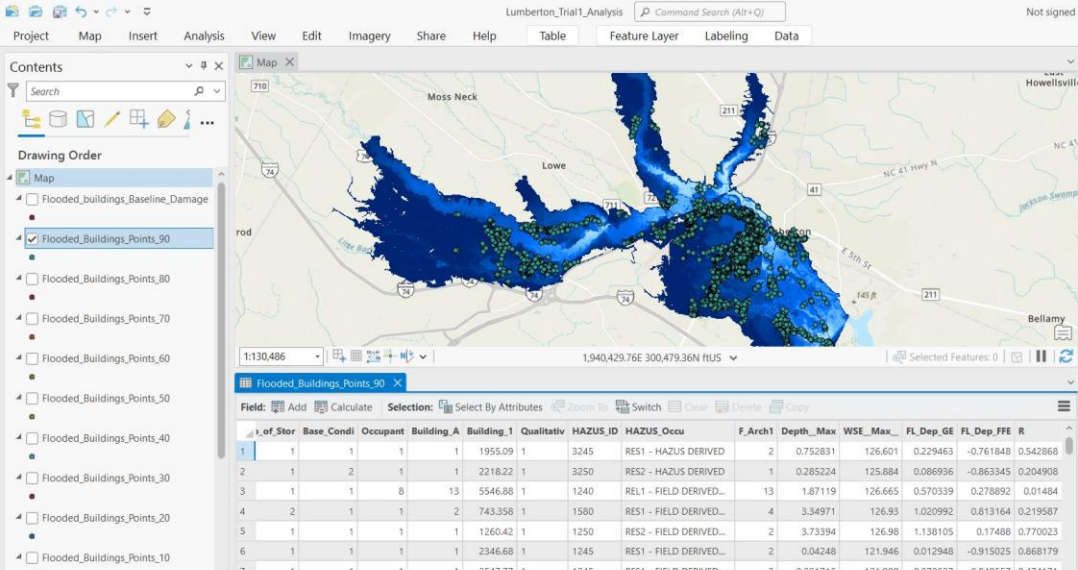
Building state conditions in different Damage States



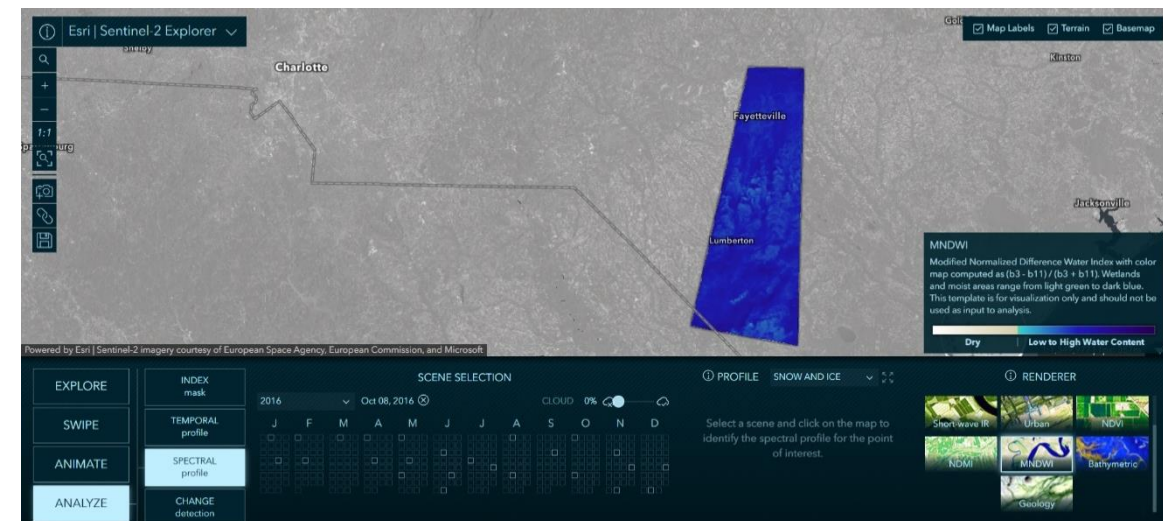
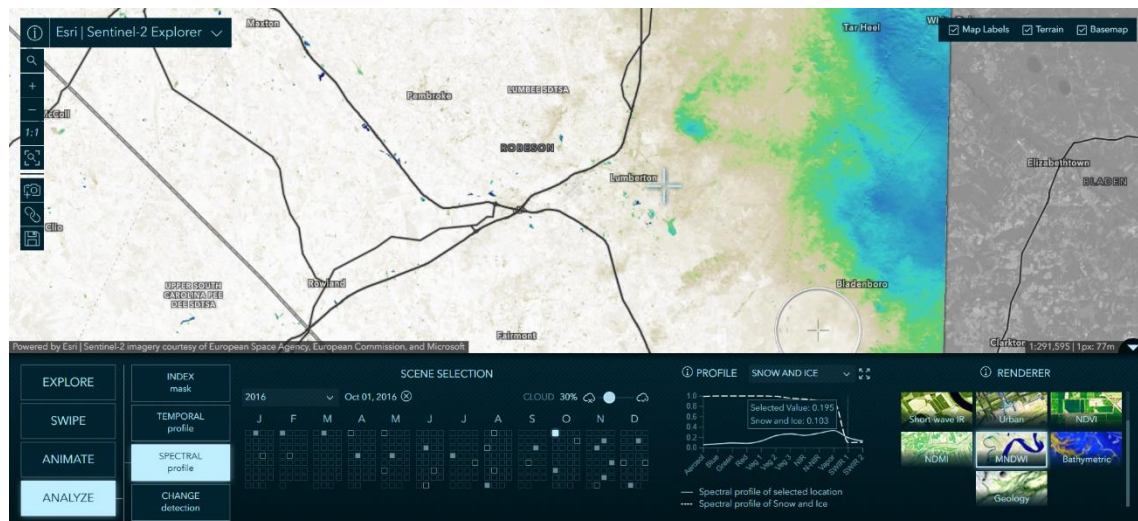
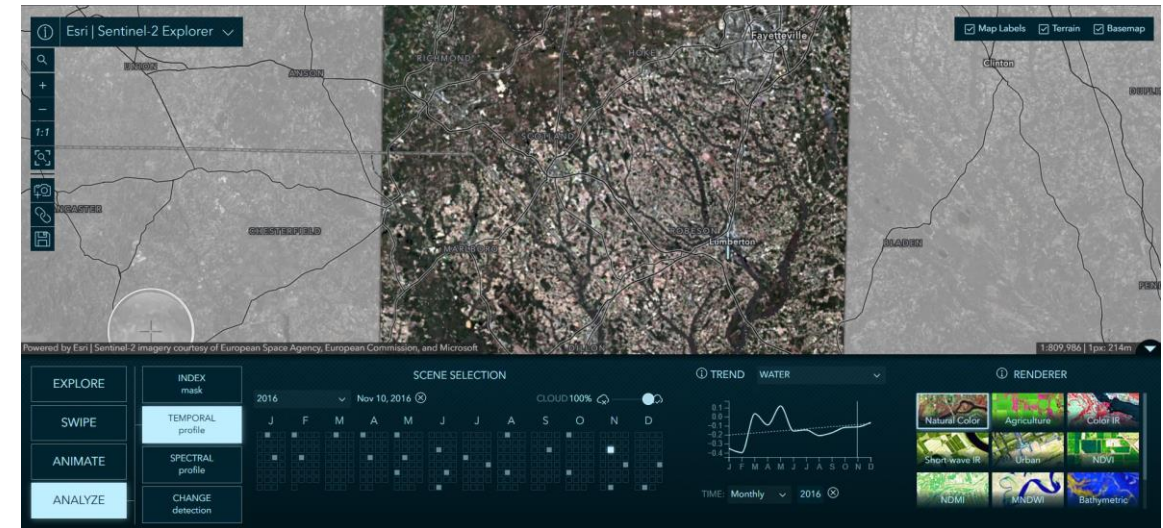
Building state conditions in different Damage States



Building state conditions in different Damage States



Analysis of flood zone in Lumberton, NC



<https://livingatlas.arcgis.com/sentinel1explorer/#mapCenter=-79.02811%2C34.62230%2C14.744&mode=analysis&mainScene=2016-10-11%7CSWI+Colorized%7C149270&sentinel1=Ascending%7CVV&tool=trend>

Since Hurricane Matthew made landfall in 2016, the Public Schools of Robeson County have been trying to recover with available resources. However, with parts of the school district still operating in temporary locations, post-disaster financial resources still being sought, and with students and staff still striving to reach stability, many challenges remain. First, at the time of Wave 2, the school district was still operating out of temporary offices. Although district officials had been actively working to identify a new location to build a permanent district office, one representative explained the challenges they were facing:

There's a lot of confusion about redoing our admin. You know, several different sides. Another kicker is when you're looking at trying to stay centrally located... when you look at Lumberton, the bulk of it is in a flood zone, so how do you plan for that, trying to acquire property or find a suitable building?

Interviewees also expressed that the absence of a permanent office was difficult for them and made their work lives more challenging, as another district official said:

You don't have a home, because work is your home away from home. And you want to be comfortable in your surroundings. You're not when you're in a temporary location. You don't know how long you're going to be here or where you're going.



Relied on probabilistic vulnerability, which propagates uncertainties regarding buildings' capacities to withstand flood depths and durations.



Predicted flooded structures and quantifies flood losses while also assessing various mitigation strategies for each of the 15 building archetypes examined in Lumberton, North Carolina.



Revealed the influence of minor alterations at the building level on the community-level flood loss estimation.



Predicted the effects of employing flood barriers, water pumps, raised structures, and their combinations in mitigating flood damage to buildings.

*Thanks
Questions or Comments?*

